Pipeline-variant resilience campaign: summary of results so far

campaign driver (Claude), for R. Canetti

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Data as of 2026-11-01 13:25 UTC. Server tree `~/variants` on machine1; status tables regenerate every 10 minutes; this document freezes them.

1 Question and arms

Starting point: the preferred pipeline = stage 1 (seeded sandwich around a random 64-bit, 1000-gate g57 source C), stage 2 three-mask modified (random helpers, unique cover), stages 3-5 as is (DB mixing 5.2 effs with the frozen identity store; split; crossing to 2x), stage 6 adaptive C-aware resynthesis v2 iterated until two consecutive rounds compress jointly by less than 2%. One fresh 40-bit seed from the OS CSPRNG is shared by all arms (stage seeds SEED+2/+3), so arms 1-3 and 5-8 share one random C and one sandwich (2,832 gates), and the balanced arm shares the seed but uses a fresh balanced C (every internal value read exactly twice; built by a port of the Codex sampler that reproduces RC's published balanced C gate for gate).

arm	description	stage 2	stage 3	stage 4	stage 5	stage-6 input
base	stage 1 as is, stage 2 three-mask modified, stages 3-5, stage 6 v2 (2% rule).	504,163	1,064,981	1,889,509	3,497,695	stage5 3,497,695
skip3	as base without stage 3. Stage 4 (split) acts on g57-form gates that stage 3 produces; on the raw stage-2 circuit it found 1,024 and grew the circuit by 0.4% in 6 s, so this arm is effectively "stage 2 -> stage 5 -> stage 6".	504,163	-	506,006	952,774	stage5 952,774
skip5	as base without stage 5: stage 6 runs on base's stage-4 output (base's stage-3/4 outputs reused verbatim).	504,163	1,064,981	1,889,509	-	stage4 1,889,509
bal	as base with a fresh balanced C in stage 1 (884 deeper internal births instead of 886).	503,865	1,062,178	1,883,952	3,487,126	stage5 3,487,126
k24	as base with stage 5 at –k-max 24 instead of 12 (base's stage-3/4 outputs and stage-4 state reused). The wider cap produced only 19 gates above 12 controls (max 14); the ANF/adversary/payload tools cap input gates at 12 controls, so this arm runs on private MAXK=24 builds of those crates.	504,163	1,064,981	1,889,509	3,497,310	stage5 3,497,310
rust	base's stage 1 + the Rust port's stage 2 (session "Rust version with circuit/stage updates", commit 5b18bff3): three-mask modified plus per-wire band turnover (promotion cohorts of 16) and fire interleaving whose measured effect is small (mean fire displacement 31 gates); 502,755 gates.	502,755	1,042,261	1,872,736	3,466,788	stage5 3,466,788
rust2	as rust with the interleaving made effective (read-only mask refresh, dirty helpers from 8 per-fire lanes of a 32-wire reserve, persistent carrier migration only every 8th fire): mean displacement 344 gates, a 250-gate window spans 3,101 original positions, 91 rotations instead of 112; 455,884 gates (commit 4a5fe63d). Both Rust arms use base's C and sandwich (identical hashes).	455,884	1,014,079	1,874,988	3,462,940	stage5 3,462,940
rust2_skip3	rust2's stage 2 straight into stages 4-5 (no stage 3), then stage 6; like skip3, stage 4 is a near no-op and stage 6 finds nothing to resynthesize.	455,884	-	457,631	853,767	stage5 853,767

2 What was measured (the columns)

- **Stage 6 v2 (C-aware)**: each round plans blocks that avoid cutting through exposed C values, resynthesizes every block from its ANF with two seeds and keeps a replacement only if it is smaller and not leakier; rounds

repeat until two consecutive rounds compress jointly by less than 2%. The smallest round output is F, the delivered circuit. Its *census* (validated: every witness re-checked on 8,192 fresh plus structured inputs) counts the deeper internal C births (886 of 936 for the random C, 884 for the balanced C; the 50-52 excluded are degree-2 functions of public inputs or outputs) present directly (verbatim up to NOT at some cut) / as an affine combination of simultaneously live wires at some cut, written d/a.

- **Raw ledger on F**: C-blind observers on F itself (direct ledger plus affine-window observers at ranks 10 and 12), labeled against C afterwards and audited; R/O = births found with >= 2 distinct representations (repeated) / exactly one (once).
- **Narrow portfolio on F**: ten C-blind window rewrites of F followed by the same observers: factor rewrites with 125-, 250-, 500- and 1,000-gate windows, factor 125 shifted by 63, factor 125 on a wire-permuted circuit, pairs 125, and three payload-view cases (factor 125, factor 250, delta 250); the 500/1,000-gate windows were added in this campaign because a stage-5 fire spans ~1,200 gates, and 2,000/4,000-gate windows were added later ("+2" in the count). The same portfolio was run on the shortened circuits, and on the 5.2-eff remixed circuit before and after its resynthesis for some arms.
- **Blind direct**: C-unaware adaptive resynthesis of F (same machinery as stage 6 with an empty census, one seed, keep-smaller, iterated to a 1% two-round rule, the previous campaign's attacker rule), with the validated census of its plateau P.
- **Remix 5.2 / remix 2**: stage-3 remixing of F with the frozen store for 5.2 or 2 effective units of work (the circuit grows about 2.1x / 1.7x), then the blind resynthesis of the remix with its census.
- **Shorten**: extra C-blind compression of P (fcompress transport compression over function-preserving reorderings, a 128-gate-cap blind pass, fcompress, a 48-gate-cap pass, fcompress; verified against P and the payload contract) with the validated census of the result; motivated by the finding that blind plateaus are not size floors.
- **Candidates C/all**: a C-blind scan on 8,192 honest samples counts the internal functions born at least twice (up to NOT) that are not affine in the inputs, then labels them against C: "deeper C births among them / all such candidates".
- **Degree-2 replicas mid-circuit**: at interior cuts (depths 0.40, 0.45, 0.50, 0.55, 0.60), how many of the 64 payload outputs are degree <= 2 polynomials of the wires live at the cut (C-blind; the degree-2 span is kept below the 36,864 samples so the test is non-vacuous). The direct payload replicas near the output (54-56 of 64 in F for most arms) are the sandwich's own copies and are not reported.
- **Sandwich classes** (rust family): the census run against the whole 128-wire sandwich as source (every gate with target < 64), classifying each sandwich gate as C g57 (936 internal, C's outputs excluded), D g57 (951), C/D transition CNOT (the 64 copy CNOTs carry C's outputs; counted only when a replica interval starts before 90% of the circuit), slicing CNOT (262), slicing Toffoli (491).
- **Tally**: deeper C births exposed in any circuit produced or measured since the campaign started on F (raw and narrow ledgers of F and of its derived circuits, every blind round output, both remixes and their rounds); strong = >= 2 supports in at least one circuit. No tallied birth is exposed only in the last 10% of a circuit (checked).
- **Inversion**: C-blind attempts to compute a preimage of the all-zero payload: a greedy structural walk (backward from the payload outputs, forward from the inputs; a meet would reconstruct C) and a CryptoMiniSat preimage search with a 24 h wall cap (8 threads), each on F and on P.

3 Results

3.1 Sizes and the C-aware stage 6

arm	stage-6 input	rounds	cumulative ratio per round	F gates	F census d/a
base	stage5 3,497,695	7	0.923/0.856/0.824/0.804 0.790/0.783/0.778	2,720,511	0/0
skip3	stage5 952,774	2	1.000/1.000	952,498	0/0
skip5	stage4 1,889,509	4	0.912/0.895/0.889/0.884	1,670,132	0/0
bal	stage5 3,487,126	7	0.926/0.862/0.833/0.813 0.800/0.793/0.786	2,742,554	0/0
k24	stage5 3,497,310	7	0.926/0.857/0.830/0.809 0.797/0.788/0.782	2,735,148	0/0

arm	stage-6 input	rounds	cumulative ratio per round	F gates	F census d/a
rust	stage5 3,466,788	11	0.905/0.805/0.752/0.717 0.696/0.679/0.668/0.659 0.652/0.644/0.641	2,221,174	0/0
rust2	stage5 3,462,940	8	0.972/0.942/0.922/0.905 0.895/0.882/0.873/0.866	2,997,188	0/0
rust2_skip3	stage5 853,767	2	1.000/1.000	853,767	0/0

3.2 Exposure of F to the C-blind observers

arm	raw R/O	narrow union R/O (cases)	factor 1000	factor 2000	factor 4000	candidates C/all on F
base	0/0	254/61 (10/10+2)	231/180	232/181	232/181	0/205,622
skip3	0/0	361/29 (10/10+1)	290/258	290/257	-	0/10,063
skip5	0/0	343/69 (10/10+2)	345/282	350/286	348/284	0/119,501
bal	0/0	295/109 (10/10+2)	282/192	283/195	279/191	0/208,884
k24	0/0	229/79 (10/10)	224/164	-	-	0/205,859
rust	0/0	322/65 (10/10+2)	281/215	284/216	284/215	0/190,181
rust2	0/0	101/27 (10/10+2)	81/52	85/60	85/60	0/165,649
rust2_skip3	0/0	42/9 (10/10+2)	44/38	44/38	44/38	0/8,873

3.3 C-unaware compression lines

arm	blind direct: rounds, ratios -> P gates, d/a	remix 5.2: gates (x F); blind of the remix	remix 2: gates (x F); blind of the remix	shorten: final (x P), d/a
base	4r 0.992/0.983/0.978/0.974 -> 2,650,806 9/9	5,549,367 (2.040) blind 12r 0.691/0.640/0.619/0.606 0.596/0.590/0.583/0.577 0.573/0.568/0.565/0.563 -> 3,125,171 37/37	4,573,901 (1.681) blind 13r 0.800/0.744/0.719/0.703 0.692/0.686/0.680/0.675 0.670/0.667/0.662/0.660 0.657 -> 3,003,030 23/23	1,786,769 (0.674) 9/9
skip3	2r 1.000/1.000 -> 952,168 0/0	1,975,704 (2.074) blind 7r 0.970/0.925/0.913/0.905 0.898/0.894/0.891 -> 1,761,278 7/7	1,652,357 (1.735) blind 4r 0.986/0.965/0.960/0.955 -> 1,578,452 1/1	539,812 (0.567) 0/0
skip5	3r 0.991/0.986/0.982 -> 1,640,742 2/2	3,567,260 (2.136) blind 9r 0.633/0.610/0.601/0.593 0.588/0.585/0.581/0.578 0.576 -> 2,056,153 16/16	2,939,382 (1.760) blind 8r 0.740/0.714/0.699/0.690 0.685/0.680/0.677/0.675 -> 1,983,686 21/21	944,984 (0.576) 2/2
bal	6r 0.990/0.981/0.973/0.964 0.961/0.956 -> 2,622,475 14/14	6,039,148 (2.202) blind 11r 0.643/0.595/0.573/0.555 0.544/0.538/0.532/0.528 0.525/0.522/0.520 -> 3,139,876 33/33	4,399,457 (1.604) blind 12r 0.815/0.760/0.731/0.715 0.701/0.687/0.679/0.675 0.670/0.666/0.663/0.661 -> 2,905,958 29/29	1,797,939 (0.686) 15/15
k24	8r 0.990/0.980/0.974/0.969 0.962/0.955/0.950/0.946 -> 2,587,709 9/9	6,042,867 (2.209) blind 10r 0.646/0.596/0.571/0.556 0.547/0.540/0.533/0.529 0.527/0.525 -> 3,173,128 36/36	4,597,623 (1.681) blind 10r 0.791/0.737/0.710/0.690 0.676/0.668/0.662/0.657 0.653/0.651 -> 2,993,584 22/22	1,806,170 (0.698) 8/8
rust	7r 0.989/0.976/0.966/0.958 0.947/0.943/0.938 -> 2,083,769 20/20	4,587,484 (2.065) blind 13r 0.665/0.608/0.584/0.572 0.563/0.557/0.550/0.543 0.539/0.536/0.532/0.529 0.526 -> 2,414,137 37/37 (running)	3,703,893 (1.668) blind 13r 0.780/0.727/0.696/0.677 0.665/0.655/0.648/0.643 0.638/0.631/0.627/0.624 0.621 -> 2,300,403 48/48	1,570,496 (0.754) 21/21
rust2	4r 0.992/0.986/0.980/0.976 -> 2,924,806 8/8	6,333,844 (2.113) blind 11r 0.697/0.640/0.614/0.594 0.579/0.567/0.556/0.548 0.541/0.536/0.531 -> 3,361,128 16/16 (running)	4,884,323 (1.630) blind 13r 0.815/0.760/0.733/0.712 0.698/0.687/0.677/0.670 0.663/0.655/0.650/0.645 0.642 -> 3,135,322 24/24 (running)	1,559,530 (0.533) 8/8
rust2_skip3	2r 1.000/1.000 -> 853,767 0/0	1,783,803 (2.089) blind 2r 0.996/0.995 -> 1,774,087 0/0	1,476,377 (1.729) blind -	476,870 (0.559) 0/0

3.4 Narrow portfolio on the derived circuits, tally, inversion

arm	**narrow on shortened**	**narrow on remix 5.2**	**narrow on remix-5.2 final**	**tally deeper (strong)**	**inversion**
base	251/79 (9/10)	106/52 (10/10)	284/91 (10/10)	492 (443)	recon_F: no meet sat_F: timeout recon_P: no meet sat_P: timeout
skip3	498/17 (10/10)	-	-	532 (515)	recon_F: no meet sat_F: timeout recon_P: no meet sat_P: timeout
skip5	339/72 (10/10)	-	-	491 (440)	recon_F: no meet sat_F: timeout recon_P: no meet sat_P: timeout
bal	331/139 (10/10)	-	-	603 (483)	recon_F: no meet sat_F: timeout recon_P: no meet sat_P: timeout
k24	246/96 (10/10)	-	-	432 (351)	recon_F: no meet sat_F: timeout recon_P: no meet sat_P: timeout
rust	294/66 (9/10)	-	-	455 (414) (running)	recon_F: no meet sat_F: timeout recon_P: no meet sat_P: timeout
rust2	129/47 (10/10+1)	29/19 (10/10)	-	234 (194) (running)	recon_F: no meet sat_F: timeout recon_P: no meet (running)
rust2_skip3	285/38 (10/10)	5/4 (10/10)	9/5 (10/10)	336 (300) (running)	recon_F: no meet (running)

Inversion detail: the walks recover 6-41 forward gates and 0-2 backward gates and never meet; every SAT run hit its 24 h cap without a result (rust2 and rust2_skip3 still running).

3.5 Exposure by test and circuit (deeper C births found / strong)

test / circuit	base F	base short	base remix5.2	base remix5.2 final	skip3 F	skip3 short	skip5 F	skip5 short	bal F	bal short	k24 F	k24 short	rust F	rust short	rust2 F	rust2 short	rust2 remix5.2	rust2__skip3 F	rust2__skip3 short	rust2__skip3 remix5.2	rust2__skip3 remix5.2 final
factor 125	67/41	56/35	2/0	117/79	60/48	277/231	39/21	67/39	60/31	54/29	68/34	62/37	110/70	83/50	26/18	21/16	1/1	2/0	108/85	0/0	0/0
factor 125 shifted 63	60/29	49/25	1/1	126/85	62/49	257/215	40/22	70/46	55/27	60/34	61/31	58/27	120/82	78/53	20/12	19/11	0/0	2/2	104/77	0/0	1/0
factor 125 wire-permuted	88/48	64/36	3/1	150/93	66/53	281/241	51/22	84/44	81/46	62/38	84/47	72/38	132/90	90/56	34/23	25/19	0/0	3/1	113/86	0/0	0/0
factor 250	114/71	-	12/4	187/121	252/221	374/350	111/73	215/149	113/61	149/79	121/51	148/92	193/126	-	45/26	53/36	1/1	12/9	194/166	1/0	1/0
factor 500	170/118	231/157	51/28	246/164	289/257	403/382	262/206	320/253	210/124	314/203	176/122	232/153	237/167	249/170	67/44	111/78	16/9	44/38	229/200	7/5	11/7
factor 1000	231/180	277/201	131/86	295/208	290/258	403/382	345/282	340/277	282/192	371/253	224/164	257/175	281/215	308/237	81/52	131/90	39/22	44/38	229/200	8/5	12/9
factor 2000	232/181	-	-	-	290/257	-	350/286	-	283/195	-	-	-	284/216	-	85/60	132/90	-	44/38	-	-	-
factor 4000	232/181	-	-	-	-	-	348/284	-	279/191	-	-	-	284/215	-	85/60	-	-	44/38	-	-	-
pairs 125	1/1	10/10	0/0	39/39	0/0	1/1	0/0	3/3	1/1	15/15	0/0	8/8	2/2	24/24	0/0	8/8	1/1	0/0	0/0	0/0	0/0
payload factor 125	74/27	47/18	1/1	150/76	50/34	214/182	61/24	67/35	90/23	58/28	70/31	64/29	136/69	91/58	22/12	25/18	0/0	2/0	94/77	0/0	0/0
payload factor 250	130/64	126/73	21/6	209/106	198/169	306/285	145/71	168/112	134/57	133/76	113/58	135/66	201/113	164/92	39/16	43/29	9/4	7/6	147/129	0/0	0/0
payload delta 250	149/79	127/71	36/13	222/114	200/170	308/290	159/75	170/110	156/67	137/77	130/65	138/65	207/121	166/93	43/23	46/32	12/5	9/8	149/128	0/0	0/0
narrow portfolio (union)	315/254	330/251	158/106	375/284	390/361	515/498	412/343	411/339	404/295	470/331	308/229	342/246	387/322	360/294	128/101	176/129	48/29	51/42	323/285	9/5	14/9
raw ledger (no rewrite)	0/0	9/5	0/0	37/11	0/0	0/0	0/0	2/0	0/0	15/4	0/0	8/4	0/0	21/12	0/0	8/5	0/0	0/0	0/0	0/0	0/0
blind direct resynthesis (all rounds)	9/9	n/a	n/a	n/a	0/0	n/a	2/2	n/a	17/17	n/a	13/13	n/a	22/22	n/a	8/8	n/a	n/a	0/0	n/a	n/a	n/a
blind resynthesis of the 5.2-eff remix	41/41	n/a	n/a	n/a	7/7	n/a	27/27	n/a	42/42	n/a	44/43	n/a	47/47	n/a	23/23	n/a	n/a	0/0	n/a	n/a	n/a
blind resynthesis of the 2-eff remix	27/27	n/a	n/a	n/a	1/1	n/a	26/26	n/a	42/42	n/a	35/35	n/a	61/61	n/a	29/29	n/a	n/a	n/a	n/a	n/a	n/a
census of the shortened circuit	n/a	9/9	n/a	n/a	n/a	0/0	n/a	2/2	n/a	15/15	n/a	8/8	n/a	21/21	n/a	8/8	n/a	n/a	0/0	n/a	n/a
TOTAL union of all tests on this circuit	334/283	330/251	158/106	375/284	393/364	515/498	419/358	411/339	435/336	470/331	329/264	342/246	407/350	360/294	159/135	176/129	48/29	51/42	323/285	9/5	14/9

3.6 Sandwich gate classes (rust family; direct/affine exposed of internal values in the class)

gate class / circuit	rust F	rust short	rust2 F	rust2 short	rust2_skip3 F	rust2_skip3 short
C g57	2/3 of 936	24/24 of 936	0/1 of 936	8/8 of 936	0/1 of 936	0/0 of 936
D g57	21/25 of 951	30/33 of 951	9/12 of 951	11/14 of 951	0/3 of 951	0/3 of 951
C/D transition CNOT	7/8 of 64	9/10 of 64	4/4 of 64	5/5 of 64	0/0 of 64	0/0 of 64
slicing CNOT	23/26 of 262	31/34 of 262	22/26 of 262	23/27 of 262	21/25 of 262	21/25 of 262
slicing Toffoli	43/43 of 491	53/53 of 491	38/38 of 491	40/40 of 491	36/36 of 491	36/36 of 491

3.7 Degree-2 replicas at interior cuts

Every arm, every circuit (F, P, shortened), every tested depth (0.40-0.60): 0 of 64 payload outputs lie in the degree-2 span of the live wires (wire rank 256, span dimension 32,897 of 36,864 samples, non-vacuous). The old baseline pipeline (managed stage 2, as-is stage 6) had 36 of 64 on F and 50 of 64 on its blind plateau at depth 0.45-0.48. With the three-mask stage 2 the band carries the payload behind cubic masks, so no interior cut exposes the outputs at degree 2, and neither the blind resynthesis nor the shortening recreates such a cut.

3.8 Gates with more than two controls

Stage 5 creates products up to the k-max and the ANF synthesizer only replaces the windows it resynthesizes; every circuit keeps wide gates except skip5 (no stage 5): base F 322,149 of 2.72M (max 12), bal 344,725, k24 338,654 (max 13), rust 123,985, rust2 936,164 (31% of its F), skip3 603,917 of 0.95M, rust2_skip3 571,393 of 0.85M, skip5 1,335 (max 5). Shortening removes most of them (about 1% of the circuit for the stage-6-compressed arms).

3.9 Block-cap experiment (does stage 6 benefit from larger blocks or more search?)

Span = gates 0-218,605 of base's stage-5 circuit (its first round-1 planner segment), two seeds, keep smaller per block, initial pass.

arm	term cap	pair cap	blocks	block gates median / p90 / max	ANF terms median / p90	plan s	orig -> best gates	ratio	ok s1/s2	timeouts	crashes	wall h	peak RSS GB
control	131072	4000000	232	887 / 1489 / 2862	17761.0 / 76685	-	218605 -> 198103	0.9062	207/208	45	0	-	-
x05	65536	2000000	262	780 / 1339 / 2865	10431.0 / 42214	204	218605 -> 199987	0.9148	238/239	43	0	0.83	1.8
x1	131072	4000000	232	887 / 1489 / 2862	17761.0 / 76685	740	218605 -> 198487	0.9080	209/208	44	0	1.28	3.4
x2	262144	8000000	207	1000 / 1674 / 4791	28304 / 155321	1110	218605 -> 200757	0.9184	194/187	32	0	2.83	4.1
x4	524288	16000000	195	1036 / 1837 / 4791	45131 / 277108	2971	218605 -> 200777	0.9184	179/179	29	0	3.44	11.6

Compression per gate is flat (0.906-0.918) across an 8x range of planner ANF budgets; the production cap is at the optimum; monomial-scaled time budgets up to 480 s change nothing (the same ~22 windows per seed time out). What compresses is ANF sparsity: ratio 0.80 below 4k terms, 0.87 at 4-16k, 0.95 at 16-65k, 0.98 at 65-131k; the windows larger budgets add (131k-524k terms) return at 0.95-0.98 after 600-1,000 s each. Per-round anatomy: the synthesizer takes 92-95% of thread-time, both seeds win symmetrically, refinement is 8% of the time for 27% of a late round's saving, and a crashed chunk's per-window fallback cost base's round 7 twenty extra hours. The driver was changed accordingly (one job pool over seeds and budgets, bisection on crashes, 60 s for dense windows from round 3, one seed after a round with $< 3\%$ gain, parallel candidate censuses, census reuse) for arms launched afterwards; rust2 ran its last rounds with 12 synthesizer processes.

4 Findings

- **The 1,000-gate window is the strongest test everywhere, and larger windows add nothing.** On the stage-6 outputs the 1,000-gate factor rewrite finds 224-345 deeper births for the Codex-generator arms where the 125-gate cases find 40-130; 2,000 and 4,000 gates give the same counts (base 232/232, skip5 350/348, bal 283/279, rust 284/284, rust2 85/85). The earlier campaign's portfolio, which stopped at 250 gates, under-reported exposure two to three times; with the long windows, base, skip3, skip5, bal, k24 and rust all end between 329 and 435 deeper births exposed, i.e. a third to a half of C.
- **rust2 is the only design that resists the window rewrites**: 128 narrow / 159 total on F (85 in the long windows), three times fewer than every other arm, and its blind lines expose 8-29 instead of 17-61. rust (turnover alone, weak interleave) is as exposed as base (387 narrow, 407 total), so the effective fire interleaving is the element that works, not the turnover.
- **Skipping stage 3 does not reduce exposure and removes stage 6's leverage**: without stage 3's identities the stage-6 synthesizer cannot compress (skip3 and rust2_skip3 plateau at their stage-5 size with 0/0 census), the narrow portfolio exposes 390 (skip3) and 51 on F (rust2_skip3), and the shortened circuits are the most exposed of all (515 and 323). Skipping stage 5 (skip5) halves the circuit but leaks like base (412 narrow, 419 total).
- **The balanced C does not help against the C-blind observers** (bal: 404 narrow, 435 total, the highest of the Codex-generator arms), consistent with the per-read exposure model of the previous campaign.
- **k-max 24 changes nothing**: stage 5 produced 19 gates above 12 controls and k24 tracks base within 1% at every stage (F 2,735,148 vs 2,720,511; narrow 308 vs 315).
- **Compression attacks expose little by themselves but shortening and remixing move the picture**: the blind direct plateaus expose 0-22 births (census) and are not size floors; fcompress-plus-small-window shortening compresses P by 25-47% and the shortened circuits are more exposed to the window rewrites than F for every arm; remixing then recompressing exposes 23-61 (census) and, for base, 375 births through the narrow portfolio of the remix final versus 158 on the remix itself.
- **No preimage from any inversion attempt**; the C-blind candidate haystack is 9k-209k repeated non-affine functions per circuit with at most 8 deeper C births among them.
- **The payload is not replicated at degree 2 anywhere inside any circuit** (0 of 64 at all tested depths for all arms), unlike the old baseline's 36-50, which closes the midpoint-replication avenue of the preimage campaign for this stage-2 family.
- **The sandwich leaks through its slicing gates, not through C**: on rust2's F, 0 of 936 C births but 9 of 951 D births, 22 of 262 slicing CNOT outputs and 38 of 491 slicing Toffoli outputs are directly present; the same pattern holds for rust and rust2_skip3.

5 Status and caveats

Still running at the time of writing: the blind remix lines of rust and rust2, rust2_skip3's resilience (its remixes were produced on the Mac Pro against the local store after the server's store was deleted to free 299 GB; the store is re-fetched automatically when the campaign ends), the remix gauntlets (rust2 and base first), the 2,000/4,000-gate windows of the arms not yet covered, the shortened-circuit portfolios of rust and rust2_skip3, and the remaining SAT inversion runs. Caveats: one seed and one C per arm (the Rust arms share base's); the attacker rules (1% stop) and the ten standard cases are the previous campaign's for comparability; the stage-6 drivers of base, skip3, skip5, bal, k24, rust and the first rounds of rust2 ran the original driver, later rounds of rust2 and all of rust2_skip3 the revised one (identical results for bit-identical inputs except for the dense-window budget and the late single seed); the degree-2 scan tests only depths 0.40-0.60.

6 Appendix: where things are

Server `~/variants` (machine1): `run/STATUS.md` and `run/STATUS.html` (tables above, refreshed every 10 min), `run/report.json`, `run/exp_blockcap/`, `run/d2/`, `run/replicas/`, `run/sandwich_exposure/`, `run/inversion/`, `run/ledgers/`, `run/resilience/`, `run/shorten/`; drivers in `py/` and `py6/`; campaign loop `py6/variants_loops10.sh`. Local: `docs/notes/pipeline_variants_20261026.md` (design and chronology), `docs/notes/pipeline_variants_status.htm` (live mirror).